

# Bohan Yu

Research Interests: Multi-Agent Reinforcement Learning; Multi-Robot Coordination and Navigation; Computer Vision

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GitHub: [github.com/yubohann](https://github.com/yubohann) | Research Portfolio: [github.com/yubohann/MultiAgent-Robotics-Portfolio](https://github.com/yubohann/MultiAgent-Robotics-Portfolio)

## EDUCATION

Hubei University of Technology

B.Eng. in Computer Science and Technology

Wuhan, China

Sep. 2023 – Jun. 2027 (Expected)

Overall Average: 86.7/100 | Academic Rank: 7/70 (Top 10%)

Comprehensive Ranking: 1/33 (coursework, research, competitions, and extracurricular activities)

Honors: First-Class Academic Scholarship(2×, Top 3%) | Outstanding Student of the Year (2024, 2025) (Top 3%)

Relevant Coursework: Artificial Intelligence (Zhejiang University Micro-Major Program), 98/100; Linear Algebra, 94/100; Advanced Mathematics I, 92/100; C Programming, 93/100; Operating Systems, 91/100

## HONORS

- **National First Prize** (Third Place), China Robot Competition and RoboCup China Open 2025 **Team Leader**
- **National Second Prize**, Robot Basketball Competition, ROBOCON 2025 **Team LiDAR Lead**
- **National Second Prize**, Robot Basketball Shooting Competition, ROBOCON 2025 **Team LiDAR Lead**
- **National First Prize**, Bionic Legged Robot Obstacle Challenge, ROBOCON 2025

## RESEARCH EXPERIENCE

### Multi-Robot Cooperative Exploration and Navigation

Research Lead and Independent Developer

Oct. 2025 - Present

Research focus: Multi-robot cooperative exploration and target search in partially observable 3D environments.

Methods:

**Belief and spatial representation:** POMDP-based belief-state modeling under map uncertainty; RGB-D semantic mapping; local geometric belief representation; connectivity graphs for regional task representation.

**Planning and coordination:** Frontier-based exploration; connectivity-aware task allocation with a capacitated vehicle-routing formulation and LKH; A\* and kinodynamic A\*; B-spline trajectory generation; virtual-structure formation control; coordination under communication, collision, and safety constraints.

**Learning and policy search:** Multi-agent reinforcement learning with PPO, Graph-SAC, Graph-MASAC, and constrained SAC; successor features; behavior cloning and DAgger; Quality-Diversity optimization with MAP-Elites.

**Technical Stack :** Python, C++, NumPy, PyTorch, ROS 2, Isaac Sim, Isaac Lab, PhysX, Habitat-Sim, Habitat-Lab, Gymnasium, PettingZoo, RGB-D, LiDAR, IMU, YOLO-World, Mobile-SAM

### Research Modules

#### Graph-Based Multi-Robot Planning and Formation Control — [AeroGate Graph](#)

- Built a fixed-height 2D multi-UAV environment for variable-size teams navigating static and moving gate layouts. Encoded vehicle state, route geometry, and gate layout as masked graph observations, implementing A\* route planning, virtual-structure formation, Graph-SAC for single-UAV control, and centralized-critic Graph-MASAC for team control.
- Added moving-gate dynamics, velocity and acceleration limits, segment-level collision checks, and an action shield for boundary, obstacle, inter-agent separation, and gate-corridor constraints.
- Compared learned controllers with A\*, Theta\*, RRT\*, Informed RRT\*, and heuristic planners under fixed single-UAV and multi-UAV settings across static and dynamic gate scenarios.

#### Procedural 3D Search Benchmark and Multi-Sensor Simulation — [AeroCityBench](#) and [Rivermark](#)

- Defined a procedural search benchmark for aerial-robot teams with an observation contract separating policy-visible observations from evaluator-held target truth. Built a geometry-derived inspection atlas, an evaluator for valid target observations, and a synchronized multi-sensor capture pipeline in Isaac Sim and PhysX.

#### Execution-Feedback-Driven Quality-Diversity for Cooperative Exploration — [HM3D Realised-QD](#)

- Developed a Quality-Diversity archive that updates from audited execution outcomes. It uses explored free-flight volume as the quality measure and team-behavior descriptors to retain distinct executable strategies, with execution and safety costs included in candidate selection.

### Multi-Robot Exploration and Geometry-Constrained Semantic Allocation

- Implemented a multi-robot task-allocation layer for object-goal navigation in HM3D, merging observed frontier cells into connected regions, filtering robot-region candidates through reachability and traversability checks, and computing a geometry-cost-minimal one-to-one assignment as the baseline, with the protocol spanning two-robot main experiments and three- and four-robot extensions.
- Built an observation-only navigation pipeline on HM3D in which online RGB-D observations with open-vocabulary detectors (YOLO-World, Mobile-SAM) update a shared semantic map and extract frontier regions; geometric feasibility screening precedes Fast Marching Method path execution, and target annotations remain visible only to the evaluator.
- Designed a semantic allocation layer in which a large language model ranks region preferences with explicit uncertainty and reorders only reachable, feasible assignments; a robust acceptance criterion replaces the geometric baseline only when worst-case semantic gain plus prospective information gain stays positive within path-cost and per-robot workload budgets, a stability margin keeps the baseline and skips global reassignment, and unreliable semantic evidence falls back to the

deterministic geometric allocation.

## **LiDAR-Inertial Localization and Shooting-Parameter Integration for ROBOCON Robot Basketball**

### **LiDAR and Perception Lead**

**Jun. 2024 - Oct. 2025**

**Methods & Systems:** ROS 2 Humble; C++17; Python; Linux; Point-LIO; FAST-LIO2; Livox MID-360; PCL; Eigen; TF2; Nav2; Gazebo; YOLO; RGB-D; UART; ESP-NOW; STM32.

- Integrated Point-LIO LiDAR-IMU localization on physical ROBOCON basketball robots under rapid acceleration, chassis vibration, mechanism motion, and partial occlusion in the competition field. Calibrated sensor extrinsics and the LiDAR-IMU time offset, then established mapping and fixed-map localization for navigation and shooting.
- Unified LiDAR odometry, fixed-map localization, and navigation through ROS 2 pose interfaces, checking pose freshness, map status, and pose jumps. Converted robot pose and hoop coordinates into range, heading error, and launch-speed parameters, then sent commands to STM32 over UART.
- Built an offline ballistic model with gravity, aerodynamic drag, and Magnus force. Used simulated annealing to search initial launch velocities across shot distances and used the model output to calibrate the online range-to-speed mapping.
- Bridged YOLO detections and RGB-D depth observations into ROS 2, using confidence, valid depth, temporal stability, and observation freshness as shooting gates. Integrated ESP-NOW position synchronization into the dual-robot task sequence with acknowledgements, heartbeat monitoring, timeout recovery, and emergency-stop handling.
- Built a Gazebo MID-360 simulation profile with Livox scan timing and IMU inputs, then led field integration across localization, perception, navigation, communication, UART, and STM32 control. The team received national second prizes in both Robot Basketball and Robot Basketball Shooting at the 24th China University Robot Competition (ROBOCON).

## **Uncertainty-Aware Counterfactual Belief-Graph World Model for Rule-Constrained Multi-Robot Tactics**

### **Project Lead and Core Developer**

**Mar. 2025 - Oct. 2025**

**Technical stack:** ROS 2 Jazzy; Isaac Lab; Python; PyTorch; Gymnasium; Stable-Baselines3; Nav2; robot\_localization; AprilTag; RGB-D; LiDAR; IMU

**Methods:** Object-centric belief modeling, typed interaction graphs, probabilistic ensembles, counterfactual interventions, SAC-Flow, CVaR-MPC, rule-aware shielding, centralized training and decentralized execution.

- Built a ROS 2 and Isaac Lab visual-adversarial environment for two robots with target ownership, range and line-of-sight gates, dwell-time triggers, pushable boxes, base armor blockers, and collision constraints. Encoded observations as fixed-type belief tokens, carrying forward stale beliefs with inflated covariance under occlusion or delay.
- Implemented action-conditioned typed graph dynamics and a probabilistic ensemble world model, separating object self-dynamics from interaction effects while predicting state distributions, reward, termination, visibility, presence, and rule risk.
- Designed paired interventions that move a box or remove an armor blocker, comparing predicted changes in state, reward, graph relations, and rule risk against matched simulator branches. Ranked SAC-Flow action proposals with receding-horizon CVaR-MPC by return, upper-tail rule risk, and ensemble disagreement, then sent only the first action through a rule-aware shield.
- Trained high-level controls for target selection, base-rush timing, blocking, recovery, task triggering, and risk preference with centralized training and decentralized execution; used a residual expert policy to retain rule-aware priors.
- Integrated AprilTag perception, robot\_localization, Nav2, shooter control, and task execution through ROS 2, then validated the stack on real robots. Separately evaluated a staged rule-level curriculum across increasing team sizes.

## **INDUSTRY EXPERIENCE**

Wuhan Yawei Electronic Technology Co., Ltd.

**Algorithm Engineering Intern**

**May 2026 - Jul. 2026**

- Developed an Isaac Lab environment for multi-UAV formation control and dynamic obstacle avoidance, integrating flight dynamics, obstacle scenes, sensor observations, and policy-to-flight-control interfaces.
- Designed policy observations encoding inter-UAV geometry, candidate waypoints, nearby obstacles, and local guidance cues, enabling joint consideration of formation error and collision risk.
- Trained and evaluated PPO-based formation-control policies with reward and termination terms for formation error, waypoint progress, safety distance, collision events, and boundary violations; tuned hyperparameters across evaluation scenarios.

## **LEADERSHIP AND TEAM EXPERIENCE**

**Aug. 2024 - Sep. 2025**

**Mathematical Modeling Laboratory**

**Team Lead**

- Led technical training for 67 members in MATLAB, Python, and linear algebra, covering problem formulation, model assumptions, algorithm selection, result validation, and scientific writing.
- Planned semester curricula and staged problem sets, then assigned modeling tasks across subteams.

**Oct. 2024 - Oct. 2025**

**ROBOCON Robotics Team**

**LiDAR and Perception Lead**

- Coordinated the LiDAR and perception subgroup with the chassis-control and shooting teams during autonomous-navigation integration.
- Mentored junior members in ROS 2, sensor drivers, YOLO-based detection, and field debugging.

**Aug. 2024 - Jul. 2025**

**College Media Center**

**Deputy Director**

- Coordinated outreach campaigns, event photography, and content for the college's official WeChat account, assigning tasks and reviewing submissions before publication.

**September 2025**

**Dawu Secondary Vocational and Technical School**

**Teaching Practicum**

- Taught mathematics to two senior-year classes, with 82 students in total, during a four-week teaching practicum.

## **MANUSCRIPTS IN PREPARATION**

Yu, B. (First author). Multi-Robot Exploration and Geometry-Constrained Semantic Allocation. Manuscript in preparation.